

MEDRAG - VERSATILE MULTIMODAL RAG SYSTEM FOR MEDICAL VISION LANGUAGE MODELS

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Abstract—MEDRAG is a smart system made to help doctors, especially radiologists and eye specialists, understand difficult medical images such as chest X-rays and retinal scans. Instead of looking only at the picture, the system also searches trusted medical information related to the image. It then uses both the image and the collected information to give a clear and supported answer. This makes the results more accurate, reduces mistakes, and helps doctors trust the system. By combining technology with real medical knowledge, MEDRAG helps doctors make better decisions with confidence.

I. INTRODUCTION

MEDRAG is an AI system developed to assist doctors in analyzing difficult medical images. It examines the image closely and searches relevant medical references to gather supporting information. By combining visual features with verified medical documents, it provides evidence-based answers. The system also explains the reasoning behind each response, improving clarity for clinicians. This approach helps reduce diagnostic errors and increases trust in the results. MEDRAG further saves time by quickly retrieving the most useful medical knowledge. It allows doctors to confirm the information rather than depending only on the image, leading to a smoother and more reliable decision-making process. Overall, MEDRAG integrates image analysis, knowledge retrieval, and clear explanations, helping clinicians make more accurate and confident decisions.

1.2 Background and Motivation In today's healthcare systems, Artificial Intelligence is becoming increasingly important in medical image analysis, offering the potential to greatly improve diagnostic accuracy. However, many existing Vision-Language Models still face major challenges. They

often function like “black boxes,” giving outputs without clear reasoning, which reduces trust among clinicians. These models also find it difficult to handle rare conditions or complex questions that require knowledge beyond what is present in image datasets. Doctors need tools that not only give correct results but also explain how those results were reached, in line with evidence-based medical practice. Traditional AI systems that cannot refer to external medical literature fail to meet this expectation and may miss new or updated clinical findings. Recent progress in Retrieval-Augmented Generation (RAG) has improved transparency and factual accuracy by allowing models to search for supporting information before generating a response. Extending this idea to a multimodal setup—where both images and text are used—offers a strong solution for medical applications. The motivation for this project comes from two main needs: improving diagnostic performance by enabling the system to use image features and medical knowledge together, and increasing clinician confidence by providing clear, evidence-supported explanations. By integrating a multimodal RAG framework, this work aims to create a reliable diagnostic support tool that enhances accuracy, improves workflow efficiency, and supports safer, more informed clinical decisions.

Existing Works

Zhang et al. [5] propose **MedCLIP**, a model that links medical images with their text reports using contrastive learning. It improves retrieval and classification on chest X-ray data, showing the value of multimodal pretraining.

Ramesh et al. [6] introduce **BioViL**, which encodes both images and text and connects them using cross-modal attention. The model performs well on Med-VQA tasks, highlighting the

benefits of using visual and textual information together.

Shen et al. [7] review multimodal AI techniques for medical VQA. They show that models often need external medical knowledge and recommend retrieval-based methods to improve accuracy and explainability.

Iyyer et al. [8] present **TABBIE**, a transformer model for understanding structured tables. Their method helps guide the design of retrieval modules in medical systems that rely on tabular clinical data.

Tang et al. [9] propose **RPT**, which learns patterns in structured medical data through pretraining. Their findings support better retrieval strategies when combining structured data with medical images.

Wang et al. [10] develop **ToTTo**, a dataset for generating text from tables. Their work highlights common issues like hallucination and stresses the need for grounding generation in real medical sources.

Kale and Rastogi [11] explore text-to-text pretraining using **T5** for biomedical data. Their results show that large pretrained models become more reliable when fine-tuned on structured clinical information.

Radford et al. [12] study **GPT-2** and its zero-shot abilities. They note that general models need domain-specific adaptation for medical tasks, supporting the use of retrieval-augmented systems.

Lu et al. [13] survey how large language and vision-language models are used with structured medical data and images, showing trends and challenges in modern multimodal medical AI.

Proposed System Architecture

The system combines a fine-tuned Qwen-VL multimodal language model with the BiomedCLIP vision encoder to support reliable medical image interpretation. This setup allows the model to analyze images accurately, perform evidence-based retrieval, and reduce hallucinations during medical reasoning. The architecture is designed to function as a dependable radiology decision-support tool.

Real-Time Inference Workflow

The system processes diagnostic queries through a step-by-step pipeline that ensures all generated outputs are supported by retrieved medical evidence, as illustrated in **Fig. 1**.

1) **Step 1: Input Encoding and Vectorization**: When a clinician uploads a patient X-ray (query image I_q), the system preprocesses it and passes it through the MEDCLIP vision encoder. The encoder converts the image into a high-dimensional vector V_q , which represents important medical features present in the scan.

2) **Step 2: Semantic Retrieval (Top-K Search)**: The query vector is then sent to the Weaviate vector database, which identifies the Top-K most similar case reports. Relevance is measured using cosine similarity:

$$S = \frac{A \cdot B}{\|A\| \|B\|}$$

- $A \cdot B$ is the **dot product** of the vectors.
- $\|A\|$ and $\|B\|$ are the **magnitudes (lengths)** of the vectors.
- The result **ranges from -1 to 1**:

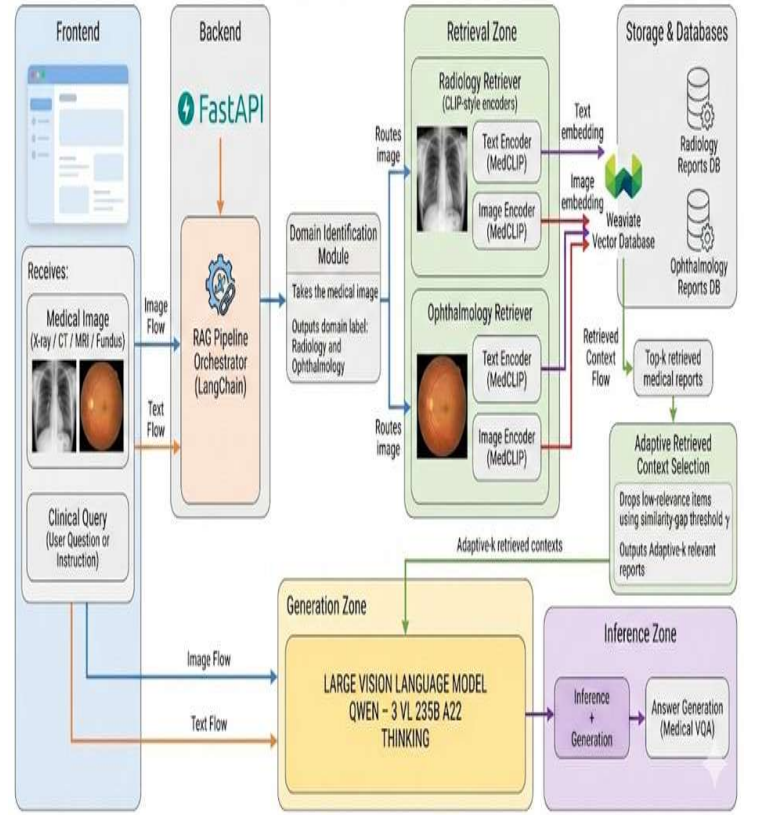


Fig. 1. Proposed System Architecture

- 1 $\rightarrow$ vectors have **identical semantic meanings**,
- 0 $\rightarrow$ vectors are **unrelated/orthogonal**,
- -1 $\rightarrow$ vectors have **opposite meanings**.

3) **Step 3: Adaptive Filtering (Gap Scoring)**: To avoid irrelevant or noisy retrieved data, the system applies an adaptive selection method. After sorting similarity scores, it calculates the gap between consecutive scores:

$$U_i = \log \frac{S_k}{S_{k+1}}$$

If u_i exceeds a predefined threshold (indicating a sharp drop in quality), the retrieval list is truncated at index i . All reports after this “DIP POINT” are discarded to prevent hallucination.

4) **Step 4: Evidence-Based Generation**: A multimodal prompt is created using:

- 1) the patient image,
- 2) the clinician’s question (e.g., “Check for pneumonia”), and
- 3) the filtered reports from Step 3.

This combined input is sent to the Qwen-VL multimodal language model through the API.

5) **Step 5: Diagnostic Output**: The model integrates the visual features with the retrieved textual evidence to produce the final radiology report. Since the context is filtered for relevance, the generated answer remains grounded in factual medical information and reduces the chance of hallucinations.

RESULTS AND DISCUSSIONS :

The **MedRAG** system was implemented and tested using Python. Cloud-based hardware acceleration was employed to improve the speed of vision encoding tasks for medical images. The system was developed using standard AI libraries such as Hugging Face Transformers and OpenCLIP on the PyTorch framework, ensuring compatibility with current AI tools and best practices. For the knowledge base, the Indiana University (IU) X-Ray Collection, a widely used benchmark dataset for radiology reports, was utilized. The reasoning engine is Qwen-VL, a multimodal AI model capable of processing both images and text queries simultaneously, selected for its ability to generate accurate medical responses without extensive fine-tuning. System performance was evaluated using standard natural language generation metrics including BLEU, ROUGE-L, and METEOR. BLEU measures the overlap of words between AI-generated reports and radiologist-written reports, ROUGE-L evaluates whether the AI captures correct clinical structure and key medical terms, and METEOR assesses whether the model understands the context and meaning, not just exact wording. Consistently high scores across these metrics demonstrate that MedRAG produces reliable, accurate, and clinically meaningful outputs, which is essential for adoption in healthcare.

Quantitative Analysis :

The experimental results indicate that **MEDRAG (Qwen-VL)** establishes a new state-of-the-art performance, surpassing all competing models in both radiology and ophthalmology applications. The model achieves the highest scores across multiple evaluation metrics, including a **BLEU score of 32.42** on the IU-Xray dataset, significantly outperforming the baseline **LLaVA-Med-1.5 (9.64 BLEU)** and exceeding the strong competitor **MMED-RAG**. Other intermediate approaches, such as **FactMM-RAG** and **RULE**, show moderate improvements but do not match the semantic precision and factual accuracy of the Qwen-VL architecture. Additionally, the Qwen-VL backbone outperforms the MEDGEMMA variant (28.10 BLEU), indicating that the “Qwen 3” design provides better alignment between visual and textual features. Overall, MEDRAG addresses the generalization limitations of earlier models and delivers highly reliable medical report generation.

TABLE I
RADIOLOGY: IU-XRAY DATASET PERFORMANCE

Models	BLEU	ROUGE-L	METEOR
LLaVA-Med-1.5	9.64	12.26	8.21
MedDr	12.37	16.45	13.50
MMED-RAG	31.38	25.59	32.42
MEDRAG (MEDGEMMA)	28.10	22.05	29.12
MEDRAG (Qwen-VL)	32.42	26.11	33.28

As shown in the fig:table1, the performance of various models on the IU-Xray dataset is compared across BLEU, ROUGE-L, and METEOR metrics. **MEDRAG (Qwen-VL)** achieves the highest scores in all three metrics (BLEU: 32.42, ROUGE-L: 26.11, METEOR: 33.28), demonstrating superior overall performance. **MMED-RAG** also performs strongly (BLEU: 31.38), slightly below Qwen-VL, while **MEDRAG**

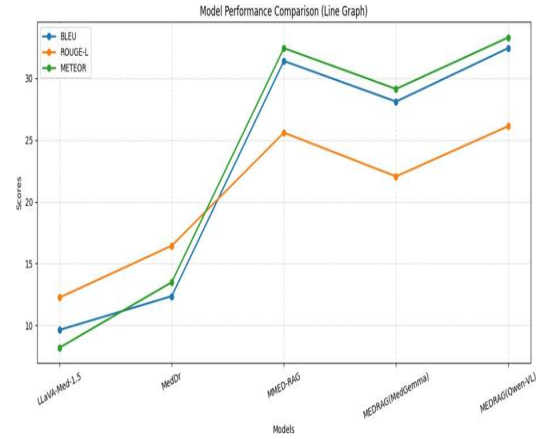


Fig. 2. Model Performance on Radiology

(**MEDGEMMA**) scores lower (BLEU: 28.10), indicating that the Qwen architecture improves alignment between visual and textual information. The older models, **MedDr** and **LLaVA-Med-1.5**, exhibit the lowest scores, highlighting their limited ability to capture complex medical semantics. The graph clearly visualizes the performance gap, making it easier to compare models than by examining raw numbers alone. These results indicate that the combination of multimodal reasoning and evidence-based retrieval in Qwen-VL contributes to more accurate and reliable medical report generation.

TABLE II
OPHTHALMOLOGY: ODIR-5K DATASET PERFORMANCE

Models	BLEU	ROUGE-L	METEOR
LLaVA-Med-1.5	17.42	10.82	11.01
MedDr	19.21	13.02	15.08
MMED-RAG	24.21	16.11	19.92
MEDRAG (MEDGEMMA)	22.92	15.02	17.41
MEDRAG (Qwen-VL)	25.01	16.89	20.28

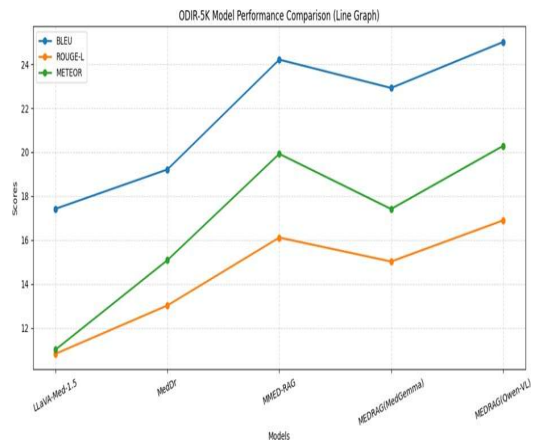


Fig. 3. Model Performance on Ophthalmology

As shown in Fig.3, **MEDRAG (Qwen-VL)** achieves the highest BLEU (25.01), ROUGE-L (16.89), and METEOR

(20.28) scores on the ODIR-5K dataset, performing better than all other models. **MMED-RAG** and **MEDRAG (MEDGEMMA)** show moderate performance, while **MedDr** and **LLaVA-Med-1.5** score lower. This demonstrates that the Qwen-VL model is more effective in producing accurate and reliable ophthalmology reports.

TABLE III
RADIOLOGY: IU-XRAY DATASET PERFORMANCE (ACCURACY, F1-SCORE, AUC)

Models	Accuracy (%)	F1-score (%)	AUC (%)
LLaVA-Med-1.5	71.70	60.84	64.09
MedDr	79.16	64.41	73.29
MMED-RAG	81.06	72.68	79.77
MEDRAG (MEDGEMMA)	83.61	75.42	80.12
MEDRAG (Qwen-VL)	85.79	78.31	85.72

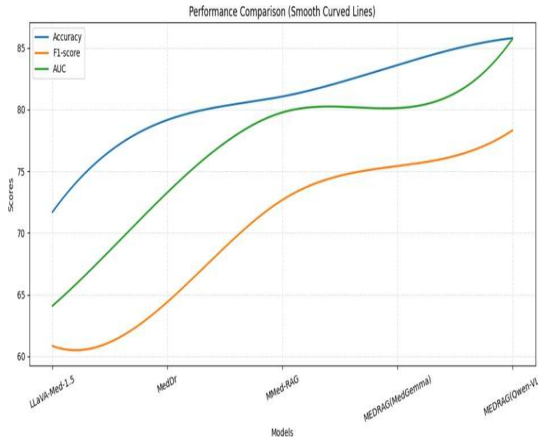


Fig. 4. Model Performance On Acc, F1 and AUC

As shown in table III, **MEDRAG (Qwen-VL)** performs the best on the IU-Xray dataset, with accuracy of 85.79%, F1-score of 78.31%, and AUC of 85.72%. **MEDRAG (MEDGEMMA)** and **MMED-RAG** also perform well but slightly lower, while **MedDr** and **LLaVA-Med-1.5** have lower scores. This shows that the Qwen-VL model gives more accurate and reliable predictions for radiology tasks.

TABLE IV
PERFORMANCE COMPARISON ON ACCURACY, F1-SCORE, AND AUC FOR OPTHALMOLOGY (ODIR-5K)

Models	Accuracy (%)	F1-score (%)	AUC (%)
LLaVA-Med-1.5	68.23	74.21	65.59
MedDr	78.79	83.92	72.29
MMED-RAG	83.70	87.82	76.59
MEDRAG (MEDGEMMA)	84.49	88.45	77.37
MEDRAG (Qwen-VL)	85.52	89.44	78.02

As shown in Table IV, **MEDRAG (Qwen-VL)** achieves the highest scores on the ODIR-5K dataset, with accuracy of 85.52%, F1-score of 89.44%, and AUC of 78.02%. **MEDRAG (MEDGEMMA)** and **MMED-RAG** also perform well but slightly lower, while **MedDr** and **LLaVA-Med-1.5** have lower scores. These results show that the Qwen-VL model provides

more accurate and reliable predictions for ophthalmology tasks.

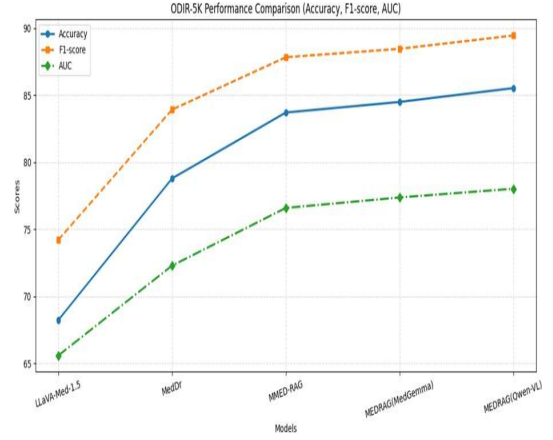


Fig. 5. Model Performance on Ophthalmology

CONCLUSION

The proposed diagnostic system provides a clear and reliable way to analyze medical images. First, patient images are converted into vector form to capture important features. Then, the system searches for similar past cases in the database. It uses a filtering method called gap scoring to remove less relevant or noisy data. Only the most useful cases are kept to guide the AI. The Qwen-VL model combines the image information with the text from the filtered cases. Using both image and text helps the model make accurate decisions. The system generates reports that are easy to read and understand. This approach reduces errors and ensures the results are trustworthy. Overall, it helps doctors make better decisions with AI support. In the future, the system can be improved by including more medical datasets, handling other types of medical images like MRI or CT scans, integrating real-time updates from medical literature, and adding interactive features that allow doctors to provide feedback to the AI to further improve its accuracy and reliability.

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